

Mikhail Razakov

Compiler / LLVM Engineer

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I work on compilers and static analysis. At MCST I implemented LLVM optimization passes and interprocedural IR analysis; I now work on SharpChecker at ISP RAS. My independent projects cover CFG/data-flow analysis, code generation, and compiler infrastructure.

EXPERIENCE

- 2026 — present

ISP RAS — Static Analysis Engineer
 - Contributing to SharpChecker, ISP RAS's static-analysis platform for C#/ .NET.
- July — August 2026

MCST — Compiler Engineering Intern
 - Implemented an LLVM LICM pass with loop analysis, side-effect/ speculative-safety checks, and hoisting of loop-invariant instructions to preheaders.
 - Developed interprocedural global-variable analysis and legality checks for safe LLVM IR transformations.

PROJECTS

- LLVM Interprocedural Global IV Optimization**

29 positive and negative regression tests, LLVM Verifier checks, idempotence checks, and before/after behavioral comparison.
- DerefAfterNullAnalyzer**

Successors are reprocessed until the state stabilizes, producing diagnostics for potentially unsafe dereference paths.
- UniversalToolchain**

Language composition is separated from execution, with architectural contracts backed by executable verification and tests.
- x86-64 Codegen & Register Allocation Lab**

A reference interpreter and differential execution check generated-code semantics; the allocator contract is verified independently.

SKILLS

- Compilers**

C++, LLVM, LLVM IR, optimization passes, interprocedural analysis, CFG, SSA, dominators, loop analysis, LICM, legality analysis
- Static / Program Analysis**

C#, .NET, Roslyn, ControlFlowGraph, fixed-point data-flow analysis, state propagation and joins
- Codegen / Tooling**

register allocation, liveness analysis, x86-64, CMake, Linux, GitHub Actions, differential testing

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

- LangDev'26 — Extensible Programming on .NET**

Accepted for an oral presentation at LangDev'26.
- UniversalToolchain — Baltic Science and Engineering Competition**

First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.
- MEPhI Junior**

Absolute winner in 2025 and 2026; in 2026, First Degree Diploma, 96/100 for UniversalToolchain.
- HSE Vysshaya Proba — Competitive & Industrial Programming**

Prize-winner in HSE Vysshaya Proba in competitive and industrial programming.
- Moscow / remote